

THE RESYNTHESIS CHANNEL: FUNDAMENTAL LIMITS OF SPEECH WATERMARKING AGAINST GENERATIVE LAUNDERING

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ABSTRACT

Audio watermarks are increasingly defeated when an attacker regenerates the signal through a vocoder or codec; this “regeneration” collapse is widely documented, yet for speech there is no theory of *which* watermarks die, under *which* attacker, or *what can survive*. We supply one. We model any blind analysis–resynthesis launderer as a *resynthesis channel* $\mathcal{W} = \mathcal{S} \circ \mathcal{A}$: an analysis map $\mathcal{A}$ that keeps only the perceptual invariants of speech (content, speaker, prosody) and a synthesis map $\mathcal{S}$ that regenerates a waveform, re-sampling everything $\mathcal{A}$ discarded. Two matched results follow. A *converse* (data-processing inequality): any watermark supported in $\ker \mathcal{A}$ —phase and psychoacoustically masked components, where imperceptibility drives post-hoc schemes—has post-laundrying detection error-exponent exactly zero. An *achievability*: a watermark aligned to the preserved invariants survives at rate R^* , the invariant sub-channel’s rate. The survivable set is exactly the non-nullspace of $\mathcal{A}$. On LibriSpeech, a phase-domain surface mark is erased by every channel; AudioSeal is defeated by a mel-spectrogram inversion *and* a neural vocoder (TPR@1%FPR 1.0 $\rightarrow$ 0.03, payload at chance) yet resists the EnCodec it was hardened against; and an invariant-aligned mark’s separability survives the vocoder and erodes gracefully under the codec. In a controlled sweep with a single fixed detector, post-laundrying detectability is monotone in the invariant-energy fraction, as the converse predicts.

Index Terms— audio watermarking, generative AI, provenance, data-processing inequality, hypothesis testing

1. INTRODUCTION

Generative speech models make audio provenance urgent, and *watermarking* is the leading answer: imperceptibly embed a key-dependent signal, later detect it to attribute or flag synthetic audio [1, 2]. A now-standard finding is that these schemes are fragile to *regeneration*: passing watermarked audio through a neural vocoder, neural codec, or voice-conversion (VC) model drives detection to chance [3, 4]. This is not incidental noise—it is the natural operation of any generative model, and it defeats watermarks that survive conventional distortions (noise, MP3, resampling).

The phenomenon is thoroughly *measured* but, for audio, not *explained*. For images this is well understood: Zhao *et al.* [5] prove regeneration removes pixel-space watermarks and prescribe embedding in preserved *semantic* content, and semantic/latent-space image watermarks (Tree-Ring [6], Gaussian Shading [7]) realize exactly that. What is missing—and what we provide—is a *quantitative theory for speech*: a matched converse and achievability that pin down not just *that* surface marks die but *how much* detectability survives

and at what *rate* R^* , exploiting the concrete, estimable invariants of audio analysis–resynthesis (content, speaker, prosody) and the phase/masking geometry specific to hearing.

Contributions. (i) We cast blind generative laundering as a *resynthesis channel* $\mathcal{W} = \mathcal{S} \circ \mathcal{A}$ (§2). (ii) A *converse* (Thm. 1): watermark energy in $\ker \mathcal{A}$ has zero surviving detection exponent—a data-processing inequality that explains the collapse, sharpened for audio by phase/masking. (iii) A matched *achievability* (Thm. 2): an invariant-aligned watermark survives at a positive rate R^* (the invariant sub-channel rate for a blind embedder); the survivable set is exactly the non-nullspace of $\mathcal{A}$. (iv) Validation on real speech (§5): across analysis–resynthesis channels of increasing nullspace (STFT and mel-spectrogram inversion, and an EnCodec codec), a phase-domain surface mark is erased everywhere, AudioSeal is operationally defeated by mel-inversion but resists the codec it was hardened against, and an invariant mark’s separability survives and erodes gracefully as the preserved subspace shrinks; in a fixed-detector sweep, post-laundrying detectability is monotone in the invariant-energy fraction, as the converse predicts.

2. THE RESYNTHESIS CHANNEL

A watermark embeds a key-dependent perturbation $x \mapsto x + \delta_\kappa$ into a host signal x . A detector that knows the key κ decides H_0 (clean, law P_0) vs. H_1 (watermarked, law P_1). The optimal Bayesian error after N i.i.d. looks decays as $e^{-N C(P_0, P_1)}$ with the *Chernoff information* $C(P_0, P_1) = \max_{s \in [0, 1]} -\log \int p_0^{1-s} p_1^s$ [8, 9]; $C = 0$ iff $P_0 = P_1$, i.e. no detector beats chance at any N .

Attacker. A blind launderer applies $\mathcal{W} = \mathcal{S} \circ \mathcal{A}$: analysis $\mathcal{A} : x \mapsto z$ extracts perceptual invariants z (a near-sufficient statistic for perceptual identity—two signals with equal z sound the same), and synthesis $\mathcal{S} : z \mapsto \hat{x}$ regenerates a waveform consistent with z , drawing everything $\mathcal{A}$ discarded from the clean prior $P_0(\cdot | z)$. The attacker does not know κ and does not optimize against the scheme (§6). Imperceptibility is a budget $\rho(\delta) \leq D$ for a psychoacoustic distortion ρ .

Linear–Gaussian surrogate. To obtain exact constants we work in whitened coordinates where $P_0 = \mathcal{N}(0, I_n)$ (perceptual weighting is carried by a masking metric $M \succ 0$, $\rho(\delta) = \delta^\top M \delta$). Let $\mathcal{A} \in \mathbb{R}^{k \times n}$ have full row rank, $P = \mathcal{A}^+ \mathcal{A}$ the orthogonal projector onto $\text{row}(\mathcal{A})$ (the *invariant subspace*), and $I - P$ the projector onto $\ker \mathcal{A}$ (the *surface subspace* $\mathcal{S}$ resamples). Then

$$\mathcal{W}(x) = Px + (I - P)w, \quad w \sim \mathcal{N}(0, I_n) \text{ fresh}, \quad (1)$$

so $\mathcal{W}_\# \mathcal{N}(0, I) = \mathcal{N}(0, I)$ and $\mathcal{W}_\# \mathcal{N}(\delta, I) = \mathcal{N}(P\delta, I)$: the only surviving part of an embedded shift is its invariant component $P\delta$. (Monotonicity holds without Gaussianity—Chernoff information obeys the DPI for any channel; the *exact-zero* for $\delta \in \ker \mathcal{A}$ uses that $\mathcal{S}$ resamples the nullspace from the clean conditional, and the surrogate supplies the constants and the achievable construction.)

Code, proofs, and data: github.com/appleweiping/resynthesis-watermark-limits.

3. CONVERSE: WHAT IS ERASED

Theorem 1 (Nullspace erasure). *For the channel (1) and any watermark δ , $C(\mathcal{W}_\#P_0, \mathcal{W}_\#P_1) = \frac{1}{8}\|P\delta\|^2 \leq \frac{1}{8}\|\delta\|^2 = C(P_0, P_1)$, with equality iff $\delta \in \text{row}(\mathcal{A})$. In particular, if $\delta \in \ker \mathcal{A}$ then $C(\mathcal{W}_\#P_0, \mathcal{W}_\#P_1) = 0$: after laundering no detector beats chance, for any N . Moreover $\mathcal{W}(x + \delta) \stackrel{d}{=} \mathcal{W}(x)$, so the laundered watermarked signal is distributed identically to a laundered clean one.*

Proof sketch. $\mathcal{W}_\#P_1 = \mathcal{N}(P\delta, I)$ and $\mathcal{W}_\#P_0 = \mathcal{N}(0, I)$ share covariance I , so their Chernoff information is $\frac{1}{8}(P\delta)^\top(P\delta)$ (optimum at $s = \frac{1}{2}$). P is an orthogonal projector, hence $\|P\delta\| \leq \|\delta\|$, giving the DPI bound, tight on $\text{row}(\mathcal{A})$. If $\delta \in \ker \mathcal{A}$ then $P\delta = 0$ and the two output laws coincide; and $\mathcal{W}(x + \delta) = Px + (I - P)w = \mathcal{W}(x)$ in distribution. $\square$

Remark 1 (Why the nullspace is huge for audio). *Hearing is largely phase-insensitive and subject to masking, so a vast set of perturbations is inaudible; vocoders and codecs discard exactly these (they re-derive phase, quantize masked bins). Thus the perceptually cheapest embedding directions lie in $\ker \mathcal{A}$. Post-hoc watermarks, optimized for imperceptibility, drift there—and die. This is the theory’s account of the empirical collapse [3, 4].*

4. ACHIEVABILITY: WHAT SURVIVES

Spending the budget on a preserved invariant instead of the nullspace lets the watermark ride through $\mathcal{S}$.

Theorem 2 (Invariant sub-channel). *Under the budget, the best surviving exponent is $\max_{\delta^\top M \delta \leq D} \frac{1}{8}\|P\delta\|^2 = \frac{D}{8}\lambda_{\max}(P, M)$, at the top generalized eigenvector of (P, M) ; it is 0 when $\delta \in \ker \mathcal{A}$. For payload, encoding a message as a surviving shift $u = P\delta \in \text{row}(\mathcal{A})$ and facing the clean host as interference ($y = u + \eta$, $\eta \sim \mathcal{N}(0, I)$) gives a blind-embedder surviving rate (informed/dirty-paper coding could exceed it)*

$$R^* = \max_{\text{tr}(M_{\text{row}}Q) \leq D} \frac{1}{2} \log \det(I + Q), \quad (2)$$

a water-filling over the eigenmodes of $M_{\text{row}} = V_{\text{row}}^\top M V_{\text{row}}$. $R^* > 0$ whenever $D > 0$; a nullspace watermark has $R^* = 0$ after $\mathcal{W}$.

Theorem 1 caps the survivable set at $\text{row}(\mathcal{A})$; Theorem 2 achieves a positive rate on it. *The survivable set is exactly the non-nullspace of $\mathcal{A}$, of size R^* .*

5. EXPERIMENTS

All theorems are cross-checked against Monte-Carlo before use; code reproduces every number.

Surrogate (validation). Fig. 1 instantiates (1) with random invariant subspaces. As a watermark drifts from the invariant subspace into $\ker \mathcal{A}$, post-laundering AUC falls from >0.88 to 0.50 (chance), matching the closed form to RMSE 0.001; the surviving detection exponent and rate R^* are positive for the invariant-aligned mark and identically zero for the post-hoc (nullspace) mark.

Real speech. We use 80 LibriSpeech test-clean utterances [10] at 16 kHz. Attackers instantiate $\mathcal{W}$ with *increasing nullspace*: a full linear-STFT Griffin–Lim [11] (a near-lossless *control*: $\ker \mathcal{A}$ is only the phase), a mel-spectrogram Griffin–Lim (lossy: $\ker \mathcal{A}$ is phase *plus* all sub-mel detail), and EnCodec [12] round-trips

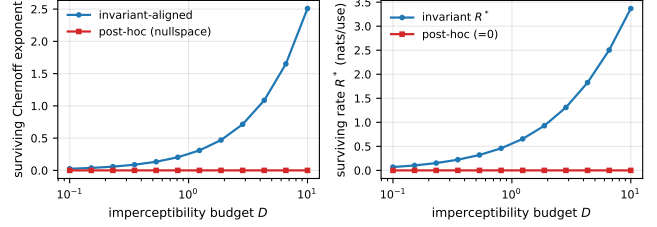


Fig. 1. Surrogate. Left: surviving Chernoff exponent vs. imperceptibility budget for the invariant-aligned vs. post-hoc (nullspace) watermark. Right: surviving payload rate R^* ; the nullspace rate is identically zero.

at 6/3/1.5 kbps (nullspace grows as the rate drops), and a *neural vocoder* (Vocos [13], mel→waveform—a learned, not iterative, resynthesizer). Watermarks: a *surface* mark (fine high-band carrier with a *waveform* detector that reads phase—in $\ker \mathcal{A}$ of every resynthesizer, whatever its magnitude energy), an *invariant* mel-envelope mark (the constructive realization of Thm. 2), and *AudioSeal* [1], a deployed neural watermark. The surface and invariant marks are at 24 dB SNR; AudioSeal at its default (27.5 dB, quieter). A mark’s mel invariant fraction f (STFT-magnitude change captured by the mel envelope) is a signal-geometry analog of $\|P\delta\|^2/\|\delta\|^2$ for a *mel-reading detector*.

Table 1 reports AUC (separability, the quantity the Chernoff exponent governs) as the nullspace grows. Survival is set by the subspace each mark’s *detector* reads. The surface mark’s waveform detector reads phase, so it is erased on *every* channel (AUC→0.5; TPR@1%FPR→0 under the lossy channels). AudioSeal is defeated by both the mel-inversion and the neural vocoder (AUC 1.00→0.152 and 0.330; TPR@1%FPR 1.00→0.03, payload 0.490 at chance under mel), while it *fully* resists the EnCodec it was hardened against (0.99/0.98/0.94 AUC). (Its sub-0.5 AUC is a sign-inversion leaving oriented separability ≈ 0.80 ; the deployed detector still fails.) The invariant mel mark—in the preserved subspace by construction—keeps separability under both the mel-inversion and the neural vocoder (0.936→0.927, 0.931; 95% CI [0.88, 0.97]) and *erodes gracefully* under EnCodec (AUC 0.74, 0.67, 0.60 at 6, 3, 1.5 kbps), the R^* behaviour of Thm. 2. Crucially, *no* mark is both operationally strong on clean audio and laundering-robust: our constructive invariant embedder is separable (AUC 0.94) but weak at 1% FPR (TPR 0.11) even clean—the very imperceptibility–robustness tension the theory predicts; a learned embedder would tighten it. Because survival is detector-dependent, cross-mark comparison is not apples-to-apples; the *controlled* test is Part B, where a *single fixed detector* sweeps a surface→invariant mixture and post-laundering AUC follows the converse’s $\sqrt{f}$ form $\Phi(a\sqrt{f} + b)$ (Fig. 2)—near chance until f clears the surface floor, then rising as the preserved invariant energy grows, as Thm. 1 predicts.

Achievability & rate. Carrying a multi-bit payload in the same invariant sub-channel (BPSK over mel-envelope patterns), the invariant mark recovers bits after both mel-inversion and the neural vocoder (bit-accuracy 0.80/0.81 at 22 dB) where a surface payload of equal size decays to chance; the surviving payload grows with the imperceptibility budget and shrinks with the attacker’s aggressiveness, the qualitative signature of the water-filling rate R^* (2) (Fig. 3).

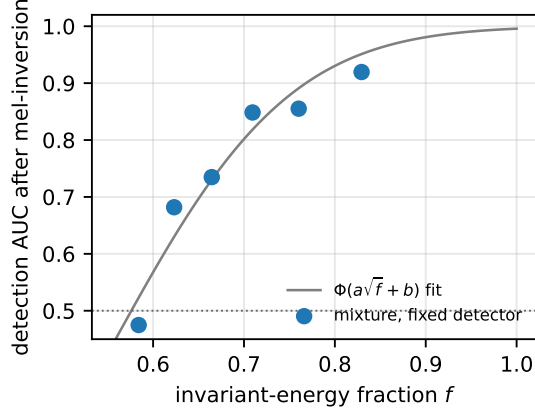


Fig. 2. Controlled converse test (mel-inversion). A *single fixed detector* reads a surface→invariant mixture; post-laundering AUC vs. the invariant-energy fraction f follows the converse’s $\sqrt{f}$ form $\Phi(a\sqrt{f} + b)$ (two-parameter fit)—near chance until f clears the surface floor, then rising. (Named marks use different detectors and are not comparable on this axis; see Table 1.)

Table 1. Real speech ($N = 80$), detection AUC *after* each blind channel (all before-AUC ≈ 1.00 except the invariant mark, 0.94). STFT[†] is a near-lossless control; mel-GL, Vocos (neural vocoder) and EnCodec (EnCk, k kbps) are lossy. The phase-domain *surface* mark is erased on every channel; *AudioSeal* resists EnCodec but is defeated by the mel-inversion and the neural vocoder (AUC 0.15/0.33, sign-inverted; TPR@1%FPR 0.04/0.05); the *invariant* mel mark’s separability survives the vocoder and erodes under low-rate EnCodec.

Watermark	STFT [†]	mel-GL	Vocos	EnC6	EnC3	EnC1.5
Surface (null)	0.45	0.54	0.49	0.53	0.54	0.56
Invariant (mel)	0.93	0.93	0.93	0.74	0.67	0.60
AudioSeal	0.98	0.15	0.33	0.99	0.98	0.94

6. DISCUSSION AND SCOPE

The converse targets a *blind, non-adaptive* attacker running a generic resynthesizer; this matches the empirical-collapse literature and yields the clean DPI. A key-aware attacker that optimizes to remove our specific mark is an arms race we do not claim to win—we treat it as a stress test, not a core result. Real analysis maps are only approximately sufficient and $\mathcal{S}$ is stochastic; our claims degrade gracefully as these idealizations break. PESQ confirms the invariant mark is near-transparent (4.61, versus AudioSeal’s 4.53), so its survival is *not* bought with audibility—at matched SNR it is in fact far less audible than the surface mark (1.99). The main remaining limitation is that we do not test *voice conversion*, a strong laundering family where the converse should bite hardest. The practical message is prescriptive: *a watermark survives generative laundering only insofar as it perturbs the invariants the generator is built to preserve*—the regime where imperceptibility is hardest, quantified by R^* .

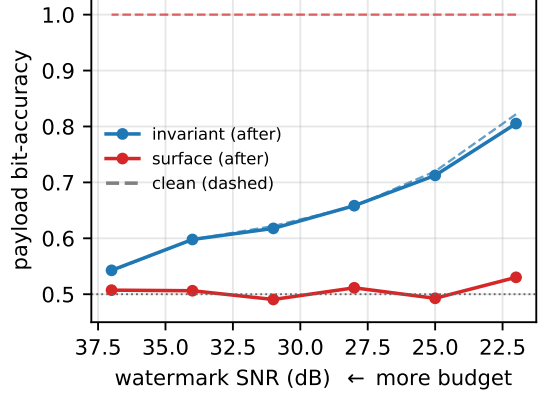


Fig. 3. Achievability. Payload bit-accuracy vs. budget (SNR): clean (dashed) vs. after mel-inversion (solid). The invariant payload *survives* (after $\approx$ clean, both modest for this proof-of-concept scheme, growing with budget); a surface payload of equal size is perfect clean but decays to chance after laundering.

7. RELATION TO PRIOR WORK

Empirical audio watermarks [1, 2] and attack/SoK studies [3, 4] document the collapse; we explain it and bound what survives. In the image domain the story is understood: Zhao *et al.* [5] prove removability and advocate semantic embedding, and Tree-Ring [6] and Gaussian Shading [7] realize regeneration-robust semantic watermarks. Our contribution is not the qualitative “embed-in-what-survives” idea but a *quantitative, provable* matched converse and achievability with an explicit surviving rate R^* for the *speech* resynthesis channel, tied to its concrete, measurable invariants. Classical information-theoretic watermarking [14, 15, 16] treats fixed additive attacks with informed embedding; our attacker is a nonlinear manifold projection and our detector is blind, locating survivability in the geometry of $\mathcal{A}$.

8. CONCLUSION

Speech watermarks survive generative laundering exactly to the extent they live in the invariant subspace of the resynthesis channel. We proved a matched converse and achievability and confirmed on spectral-inversion and neural-codec channels that detection collapses as the surviving energy vanishes. Designing watermarks *in* the invariant sub-channel, at rate R^* , is the path to laundering-robust provenance.

9. REFERENCES

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